BDA The Association of UK Dietitians
THE OFFICIAL JOURNAL OF THE BRITISH DIETETIC ASSOCIATION
Journal of Human Nutrition and Dietetics

INVITED REVIEW

Breastfeeding as a public health responsibility: a review of the evidence

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Keywords
breastfeeding, formula feeding, infant feeding, public health, society.

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How to cite this article
Brown A. (2017) Breastfeeding as a public health responsibility: a review of the evidence. *J Hum Nutr Diet*.
https://doi.org/10.1111/jhn.12496

Abstract

Background: Although intention to breastfeed in Western culture is high, many women stop breastfeeding before they are ready. From a physiological perspective, rates of primary milk insufficiency or contraindications to breastfeed should be low. However, numerous women encounter numerous barriers to breastfeeding, many of which occur at the social, cultural and political level and are therefore outside of maternal control. This review identifies and examines the impact of these barriers and considers how public health services should play a central role in creating a supportive breastfeeding environment.
Methods: A narrative review to synthesise themes in the literature was conducted, using Web of Science, PubMed and Science Direct. Barriers to breastfeeding at the societal rather than individual level were identified (e.g. in relation to health services, policies and economic factors). Only English language papers were included.
Results: Many barriers to breastfeeding exist at the societal rather than individual level. These influences are typically outside mothers' control. Five core themes were identified; the need for investment in (i) health services; (ii) population level health promotion; (iii) supporting maternal legal rights; (iv) protection of maternal wellbeing; and (v) reducing the reach of the breast milk substitute industry.
Conclusions: Although individual support is important, breastfeeding must be considered a public health issue that requires investment at a societal level. Focusing solely on solving individual issues will not lead to the cultural changes needed to normalise breastfeeding. Countries that have adopted a multicomponent public heath strategy to increase breastfeeding levels have had significant success. These strategies must be emulated more widely.

Introduction

Breastfeeding is established as protecting infant and maternal health [1,2] thus reducing healthcare costs through decreased use of services [3,4]. Global public health policy therefore recommends exclusive breastfeeding for the first 6 months of life, continued alongside solid foods for as long as mother and infant desire [5]. However, despite this, breastfeeding rates in many Western countries remain low. Only half of mothers in the USA and Australia are giving any breastmilk at all by 6 months, with only one-third doing so in the UK [6].

These figures cannot be explained by weak intentions. In the UK, over 80% of women want to breastfeed [7]. Moreover, early breastfeeding cessation can be associated with feelings of guilt and regret [8] and even post-natal depression [9]. Furthermore, low rates cannot be explained by a widespread primary physiological inability to breastfeed. Although some health issues, such as polycystic ovary syndrome and hypoplastic breasts, can impede milk production, partial or even full milk production is possible for some. A limited number of medications are contraindicated with breastfeeding, although there is often an alternative. These issues should, however, be statistically rare across populations [10].

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Instead, it is well established that maternal experiences heavily influence breastfeeding intention, initiation and duration. Aside from physical pain and difficulties, issues with stretched professional care, negative social attitudes, body image, conflicting responsibilities and a lack of familial support have all been highlighted as barriers to breastfeeding [11–13]. Unfortunately, many of these societal factors can lead to women not breastfeeding responsively, which in turn can negatively impact upon milk supply, leading to cessation [14].

These influences are notably based on social and cultural attitudes and on values wider than the individual mother, yet many interventions aiming to improve breastfeeding focus primarily on supporting women at the medical and individual level. Although these are vital, effective and valued services [15–17], tackling issues such as pain and physical difficulty once they have arisen is only part of the solution. Instead, as in many areas of health, a preventative, public health approach to enable women to breastfeed is also needed.

It is widely recognised that our behaviour as individuals is affected by the systems and structures of the environment and society in which we live [18]. Social, economic and political factors all influence our knowledge, attitudes and ability to make healthy choices [19]. Public health recognises this and puts in place systems that give individuals the best possible chance of health, seeking to promote healthier choices and reduce the risk of illness occurring [20]. Examples of this include the prohibition of smoking in public places, adding fluoride to the water supply and removing value added tax from fresh produce. A key aspect of public health is using health promotion campaigns to raise awareness and change behaviour not just for those who issues might affect, but for all [21].

Recently, attention at a policy level has turned to the importance of taking a public health approach to support breastfeeding women. In 2016, a seminal series in the *Lancet* highlighted the importance of public health to breastfeeding success emphasising:

> The reasons why women avoid or stop breastfeeding range from the medical, cultural and psychological, to physical discomfort and inconvenience. These matters are not trivial and many mothers without support turn to a bottle of formula.[22]
>
> and
>
> The success or failure of breastfeeding should not be seen solely as the responsibility of the woman. Her ability to breastfeed is very much shaped by the support and the environment in which she lives. There is a broader responsibility of governments and society to support women through policies and programmes in the community.[23]

These quotes highlight how essential a wider public health approach is to supporting breastfeeding mothers. Although it is the mother herself who ultimately breastfeeds, her ability to do so is affected by the culture and context she lives in. Rather than considering breastfeeding solely as an individual issue that the mother is responsible for solving, public health must take broader ownership by identifying how wider issues at the societal level can be changed to also enable breastfeeding. The aim of the present review is to highlight which aspects policies and programmes need to target to affect breastfeeding at the public health level, as well as the potential mechanisms to do this.

Methods

A narrative review was conducted to synthesise themes in the literature. A literature search was conducted using Web of Science, PubMed and Science Direct, specifying dates from 1997 to 2016. Search terms included individual and combinations of milk type/process (breastfeeding; breast milk; formula feeding; formula milk; artificial feeding; infant feeding; bottle-feeding), behaviour (initiation; continuation; cessation; stopping; duration; decision; choice; reason) and influence (social; cultural; economic; political; industry; environmental; health professionals; public; public health). Studies examining wider social, cultural and economic influences upon breastfeeding that could be targeted at a public health level were included, whereas research that examined physiological properties of breast milk, health impacts and physiological complications/impediments was excluded. Studies were limited to the English language. Both quantitative and qualitative studies were included because women's experiences are an important body of work in this area.

Results

Five core themes were identified; the need for investment in (i) health services provision; (ii) population level health promotion; (iii) supporting maternal legal rights; (iv) protection of maternal wellbeing; and (v) reducing the reach of the breast milk substitute industry. The importance of these within wider social, economic and cultural communities is considered.

1. Investment in health services provision

The health system in which a woman gives birth in affects her ability to breastfeed. This is not simply related to services supporting breastfeeding; maternal experiences during pregnancy, birth and post-natal care can all impact upon breastfeeding intentions and behaviour at both a physiological and psychological level.

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Antenatal education

Good quality antenatal breastfeeding education is associated with a longer breastfeeding duration [24]; however, many women report that their experience of antenatal education in relation to breastfeeding is lacking [25]. Often breastfeeding education does not prepare women to breastfeed; it highlights benefits but does not discuss the process of breastfeeding and how it may differ from formula feeding. This can lead to mothers questioning whether the behaviour of their breastfed infant in comparison to formula fed infants is wrong [26].

At the heart of providing good quality antenatal education should be an emphasis on the potential differences between breast and formula feeding. Breastfed infants feed more frequently and irregularly than formula fed infants [27], partly as a result of the faster digestion of breast milk [28]. Breastfed babies also take less milk per feed from the first day of life, leading to more frequent feeds [29]. In the early days, breastfed babies will feed more frequently at night [30], although this difference disappears by 6 months [31].

This information is not always covered in antenatal education [26], leading to concerns that breastfed infants are not receiving sufficient milk [32]. This can lead to unecessary supplementation [33] or attempts to manipulate feeding patterns, both of which can reduce milk supply [34]. Breastfed infants often lose more weight after birth and regain this weight more slowly than formula fed infants which can exacerbate concerns [35]. This difference is a common concern for breastfeeding mothers [36].

Labour and delivery

Critics have long challenged the impact of increasing medicalisation of childbirth in Western culture on maternal and infant health. Unnecessary monitoring, interventions and a lack of continuous support can lead to an increased risk of complications [37]. However, interventions during childbirth can also negatively impact on breastfeeding success. Infants born by caesarean section are less likely to be breastfed [38]. Aside from associated pain, the release of oxytocin and prolactin is weaker after a caesarean, delaying milk production [39]. Babies born by assisted delivery are less likely to be breastfed at 2 weeks, potentially as a result of bruising affecting latch [40]. Medications during labour, particularly pethidine, can also affect breastfeeding outcomes because of sub optimal rooting and latch [41]. Epidurals have also been associated with reduced breastfeeding continuation [42].

Post-natal care

High-quality post-natal care plays a critical role in breastfeeding success. Hospitals that are accredited as Baby Friendly, following the 10 steps to successful breastfeeding have higher breastfeeding rates, with the more steps a hospital follows, the better their breastfeeding outcomes [43]. Pre- and post-adoption Baby Friendly studies suggest an increase in breastfeeding rates attributed to these steps [44].

For example, infants who have skin-to-skin, placed on their mother's chest after the birth, are more likely to breastfeed, which may be explained through better latch and stronger sucking skills [45]. Keeping mother and baby together after birth is another important step. In one randomised controlled trial, only 45% of babies allocated to a hospital nursery after birth were exclusively breastfed on discharge, whereas 86% of those kept with their mother were [46]. Formula supplementation can also damage breastfeeding because it can decrease supply [47]. In one study, infants who received formula supplements in hospital were twice as likely not to be breastfeeding at 1 month compared to those who had been exclusively breastfed [48].

Change needed

Many breastfeeding issues may be preventable if women had better knowledge and enhanced support during the birth and in the post-natal period. These services may be particularly vital in neonatal care units, where infants obtain the greatest benefit from breastmilk, yet complications with breastfeeding from an infant prematurity perspective or as a result of maternal birth complications make breastfeeding more difficult [49].

Support specifically in relation to breastfeeding during the antenatal and post-natal period from well-qualified, knowledgeable staff is critical [50]. However, many women fail to receive high-quality care from the professionals who support them [51]. For many, this is not because of a lack of professional interest. Increasing demands on midwives and health visitors to care for more women, with fewer resources, leads to many professionals feeling angry and frustrated that they do not have the time to offer support. Consequently, formula can sometimes be offered as a quick solution [52]. However, not all professionals are supportive of breastfeeding; some do not value giving breastfeeding support, perceive little benefit or are reluctant to discuss breastfeeding in case it makes mums feel guilty [53].

These issues indicate the need for healthcare systems to support breastfeeding from the start. On a policy level, this involves prioritising breastfeeding, with increased resources needed for staff to spend time with new mothers during pregnancy, labour and after the birth. High-quality detailed antenatal breastfeeding education is needed that focuses on the realities of normal breastfeeding rather than aiming to increase intention alone. Investing support during the birth may impact upon breastfeeding success because fewer complications will arise [54]. More intensive breastfeeding support is needed for those mothers who have been through a complicated delivery.

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Post-natally, an increase in staffing on hospital wards and in the community is vital for ensuring that women receive the one-to-one support needed after the birth. Peer support should offer an additional level to this. Mothers value the knowledgeable and empathetic relationship that peer supporters can offer [55], particularly in areas where breastfeeding levels are low, because it provides a community that is supportive and normalising of breastfeeding [56]. However, government cuts to services have seen many of these groups close.

In addition to increasing staffing, investment must be made into increasing the training of professionals with respect to understanding breastfeeding and valuing supporting new mothers [57]. This should be implemented across the spectrum of those who interact with breastfeeding mothers, rather than simply midwives and health visitors alone. For example, providing doctors with educational training increases breastfeeding rates particularly if solutions on how to effectively manage difficulties are focused on [58].

2. Investment in societal public health messages

Numerous models of human behaviour show that affected individuals should not be the sole target of health promotion because the attitudes and behaviours of those around them affect their decisions [59]. This is the case for breastfeeding; mothers who feel supported by those around them are more likely to initiate and continue breastfeeding [60]. Unfortunately, societal understanding, value and support of breastfeeding in many Western countries is poor. Although mothers may be told 'breast is best', many think formula is sufficient, with little tangible difference between the two methods [61]. Societal experience, knowledge and understanding of infant feeding is also often heavily weighted towards bottle-feeding [62].

The attitudes and experiences of those close to the mother do matter. A strong predictor of breastfeeding is whether a woman was herself breastfed [63]. A grandmother who has experience of breastfeeding is more likely to be supportive and able to offer practical advice [64]. However, many grandmothers today fed their infants in a time when breastfeeding rates were very low and routines for feeding common [65]. This may mean that grandmothers, although supportive, may not be able to offer up to date advice [66]. Conversely, some grandmothers may try to dissuade their daughter from breastfeeding [67] and the more frequent contact a mother has with her mother in this circumstance the less likely she is to breastfeed, particularly for younger mothers [68].

Additionally, partner attitude matters. When a partner is supportive, mothers are more likely to breastfeed, particularly if they act as her advocate if she experiences difficulties [69]. Although most fathers are supportive, many feel helpless if she experiences difficulties [70], wanting more information about how to support breastfeeding, yet many are excluded from antenatal breastfeeding education sessions [71]. Conversely, when a father feels excluded, wants his partner to return to 'normal' or feels embarrassed, women are less likely to breastfeed [72].

Finally, wider public attitudes can affect maternal decisions. Around a third of the public in the UK, USA and Australia consider that a baby should not be breastfed in public [73]. This is not simply about physical encounters; many consider it inappropriate to show breastfeeding on the television or in print [74]. These attitudes are strongly tied into perceptions of the sexualisation of the female body and an assumption that breastfeeding should be a private act [75]. This is further tied to male sexist attitudes towards women. Men who score highly on sexist traits are more likely to react angrily to breastfeeding in public [76]. Breastfeeding may imply to some that a woman's priority is motherhood rather than for the purpose of being sexually available, which can incite anger in those who consider women should be sexually available to them [77].

Change needed

Interventions to improve breastfeeding should not be targeted solely at the mother. Those around her have considerable influence and need education about the importance of breastfeeding and their support role, particularly for older generations whose breastfeeding knowledge may have been learnt when infant feeding advice was markedly different [26].

Interventions that target the knowledge of fathers have been successful at increasing breastfeeding rates, particularly those that teach fathers to identify and solve breastfeeding issues [78]. Caution must be taken about the messages given because a potentially increased involvement of fathers in infant care can lead to lower breastfeeding rates [79]. Some mothers may also be uncomfortable with learning about breastfeeding in front of other men. Sensitivity is key, particularly in relation to cultural differences [80]. Likewise, involving grandmothers in education, particularly to update them on current guidelines, reduces the amount of unhelpful advice [81]. This education should be more widespread.

Additionally, given the influence of societal attitudes upon maternal confidence [7], wider public health campaigns to improve public perceptions of breastfeeding (and challenge the notion of breastfeeding as sexual) are vital. Australia, for example, has developed a series of breastfeeding adverts to promote acceptance and knowledge of breastfeeding [82]. Images of breastfeeding, and the inclusion of breastfeeding in television and other

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media could be beneficial with respect to portraying breastfeeding as normal and part of life rather than current media stereotypes of being difficult, sexual or comical [83]. Viewing images of breastfeeding mothers increases positive attitudes towards breastfeeding [84]. Social media could be a useful vehicle for accessing a wide audience, particularly amongst younger demographics [85].

These interventions should ideally start young as part of biology and social education [26]. Many teens have never been exposed to a woman breastfeeding [86] and yet hold some of the strongest negative views towards breastfeeding, or doing so in public [87]. However, teens who have witnessed breastfeeding are more likely to plan to breastfeed in the future [88]. Creating a positive cycle is needed, although breastfeeding is often missing from the school curriculum. In one study, only half of teachers considered it appropriate for primary school age to learn about breastfeeding [89]. Again, challenging unhealthy societal notions of breastfeeding as sexual is central. The key way to do this is to enhance exposure of it.

3. Policy and law to protect breastfeeding mothers

In many Western countries, various laws are in place to help protect breastfeeding mothers. However, media stories frequently illustrate how often these laws are broken, either deliberately or as a result of poor understanding, or are not sufficiently detailed to support new mothers fully.

The right to breastfeed in public

Linked to the discussion above regarding public attitudes towards breastfeeding, the maternal right to breastfeed in public must be more strongly reinforced. Mothers in the UK, Australia and many areas of the USA are protected by law to breastfeed their infants in public places. However, despite this, many breastfeeding women report receiving negative comments or even being asked to stop breastfeeding [90]. Members of the public also feel they are entitled to negative attitudes regarding breastfeeding in public, despite laws put in place. In one US study, fewer than 60% of respondents agreed that women should have the right to breastfeed in public [91]. Because of this, despite legal protection, many women naturally feel uncomfortable at breastfeeding in public places [92].

Maternity leave

Increased paid maternity leave is associated with a longer breastfeeding duration [93]. However, the amount of paid leave varies widely. In the USA, most women receive no paid leave. Conversely in the UK, there is paid leave until 9 months, although it drops to a lower amount after 6 weeks. Many women, particularly if they are the main wage earner, return to work for financial reasons when they are still breastfeeding frequently and need to make a decision to stop, or express milk during their working day [94]. Concerns around inflexibility and balancing both can lead to mothers stopping before they return [95]. This particularly affects women in less senior positions, those who are lower paid, and with inflexible jobs [96].

Return to work

Linked to the duration of maternity leave, needing to return to work is a common reason given for stopping breastfeeding, or not initiating it at all [97]. A fifth of women in the UK report this as a barrier [7] and similar patterns are seen in many Western countries [98].

Workplace policies can significantly affect whether mothers are able to breastfeed on return to work. In the USA, for example, the break time for nursing mothers law requires that employers provide a time and place for employees to express milk [99]. Conversely, in the UK, the law simply requires that organisations provide a room for breastfeeding mothers to be able to rest and lie down, although there is no requirement for women to be given paid breaks [100]. This means that many women do not have breaks, let alone paid breaks, to express, making breastfeeding maintenance a challenge [101].

If women do choose to express at work, their colleagues can react negatively, damaging maternal confidence to do so. Reactions can include embarrassment, jealousy and even offence [102]. Subconsciously, bodily fluids can be associated with inconvenience and illness and some view it as leaking sexual fluids in the work place [103]. Many women feel too intimidated to complain.

Change needed

Interventions to increase breastfeeding rates must consider how wider policy can be used to protect women from a legal standpoint to breastfeed for longer. In terms of the law supporting women to breastfeed in public, a greater awareness and understanding of why these guidelines are in place is needed. Approaches such as the Breastfeeding Welcome Scheme, where public places sign up to support breastfeeding in their venue, are important. The scheme works not only by reassuring mothers, but also sending a message to others that breastfeeding is acceptable. Large scale roll out of this in Australia had a positive impact upon maternal feelings of acceptance, alongside increased public awareness [104]. Potentially, sanctions are needed for public places (or individuals) who break this law.

In terms of maternity leave, governments must invest in extended leave that is paid at a feasible level for women to use. Countries such as Sweden, where maternity leave is

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paid at a high rate for an extended period of time, have some of the highest breastfeeding rates in the West [6]. Given the economic benefits of breastfeeding for mother, infant and society [3,4], this investment would likely pay a significant return.

Supporting mothers to return gradually to work and ensuring that work place regulations such as paid breaks are in place to support continued breastfeeding across all types of employment is important and also brings a financial return for the organisation. In the USA, employers that are deemed breastfeeding friendly (e.g. allowing breaks and providing a suitable location to express milk) have increased morale, decreased turnover and increased retention [105]. These improved outcomes, combined with lower health costs for staff health insurance, mean that investing in breastfeeding directly saves the employer money, which is estimated at a $3 saving for every $1 invested [106]. These interventions work; states that have written the right for breastfeeding breaks and private space into law have higher initiation and continuation rates [107].

4. Protection of maternal wellbeing

Infant feeding does not happen in isolation; it is part of mothering. However, the transition to motherhood can be a challenge; many mothers are not prepared for the intensity of caring for a newborn, reporting feelings of anxiety and shock and even grief [108]. Professional, educated mothers feel this shift most harshly [109]. This is exacerbated by isolation for many. In non-Western cultures, it is traditional for family to care for the mother after birth, letting her rest and recover [110], whereas new mothers in Western culture are often almost solely responsible for infant care, leaving them exhausted.

It is unsurprising that maternal identity issues, exhaustion and isolation are risk factors for post-natal depression, which in turn is associated with a decreased breastfeeding duration [111]. Confidence and self efficacy play key underlying roles here. Although wider maternal parenting confidence is associated with increased breastfeeding duration [63], post-natal depression can make the concept of breastfeeding, or infant care in general, feel difficult and overwhelming [112]. Low maternal self efficacy is associated with increased perception of insufficient milk [113].

Breastfeeding difficulties, particularly with pain and perceived insufficient milk, are associated with an increased risk of both stopping breastfeeding and post-natal depression [9]. However, feelings of depression and anxiety can interfere with the ability to breastfeed. Mothers with post-natal depression are more likely to have poorer interactions with their newborn, such as being less sensitive in their touch and positioning on the breast, which can lead to breastfeeding difficulties [114].

Change needed

Alongside enhanced investment in breastfeeding support to prevent difficulties arising, greater investment is needed more widely into supporting new mothers. Mothers need to feel socially supported in their new role, which can also help give mothers more confidence to breastfeed [115]. Professional support will play a vital role in this. Listening visits from health visitors are valued by new mothers and are associated with a clinical reduction in symptoms of post-natal depression and a rise in life satisfaction [116]. However, a lack of funding in the area means that services are stretched. Investment is needed.

Awareness of the importance of maternal mental health needs to extend to everyone, not simply those with a professional role to care. Enhancing community support for new mothers would likely reduce post-natal illness and enhance breastfeeding rates. Raymond discusses the concept of mothers building a safety net around them of community contacts that could support them when their mood is low [117]. Likewise, peer support can reduce symptoms of depression and anxiety [118], which may in turn help women to breastfeed for longer.

5. Reducing the reach of the breast milk substitute industry

Finally, despite the global promotion of breastfeeding, sales of formula milk reach over 40 billion US dollars per year [119]. Although the International Code of Marketing of Breast milk Substitutes [120] sets out a code banning the promotion of breast milk substitutes for babies under 6 months old, this only has to be followed if written into the law of individual countries. Even in countries where it becomes legislation, this regulation is often broken or there is variation regarding to what extent countries follow the code [121]. For example, although banned in the UK, free infant formula is commonly included in hospital discharge packs in the USA [122].

Companies also circumvent regulations with brand advertising. Advertisements focus on toddler milks, although brand recognition increases sales for other products in that range [123]. Many cannot tell the difference between adverts for infant and follow-on formula [124]. Mothers who recall seeing formula adverts on television are more likely to formula feed [125].

Change needed

Governments must take responsibility for ensuring that the code is met. Simple interventions such as preventing hospitals from advertising products in discharge bags increase breastfeeding [126]. Fuller implementation is also needed to close loopholes such as companies offering

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education to practitioners or pregnancy support lines for women would. Fines for breaking laws, or exaggerating claims, should be more widely used.

Simultaneously, a more active approach is needed in using similar tactics to promote breastfeeding. As noted above, this might comprise using government funded adverts that adopt the same strategies as those used by formula companies (rather than simple breast is best messages) and ensuring a greater visibility of breastfeeding in the public sphere [26].

Discussion and conclusions

The complexity of infant feeding decisions is seen in the number of systems level factors that affect maternal behaviour. Although biological impediments exist, environmental influences, particularly at the societal level, are pervasive. Maternal decisions and the ability to breastfeed are affected by the knowledge, attitudes and expectations of the society around her. To change breastfeeding, we must therefore change how breastfeeding and mothering is perceived in our society by removing structural barriers rather than targeting the individual alone [23]. We must create an environment where breastfeeding is normal, accepted and protected.

Governments must invest financially into protecting new mothers, not least because of the potential financial return. However, although some aspects such as laws and policy can easily be universal, given limited economies, interventions must focus on the most vulnerable in our society. Mothers who live in areas of economic deprivation are the least likely to breastfeed [127], yet their infants stand to benefit the most from being breastfed [128]. Poverty itself does not damage breastfeeding, but formula use is normative amongst more deprived communities [56]. Lower incomes and job insecurity may also affect ability to breastfeed dictating an earlier return to work [129] In addition, an increase in mental health issues and poorer social support can make breastfeeding more challenging [130].

A complex relationship is also seen with ethnicity. In the UK, breastfeeding rates amongst mothers from non-white backgrounds, particularly immigrants to the UK, are significantly higher than those from white backgrounds [7]. This is attributed to social norms around motherhood and feeding. However, over time, this relationship diminishes, particularly if families adopt Western norms and values surround infant feeding [131]. Conversely, in the USA, women from Black American backgrounds are significantly less likely to breastfeed. Poverty, community norms, a history of oppression and a lack of imagery of black women are key influences [132]. Black women in the USA are statistically less likely to live in an area attached to a hospital that follows the Baby Friendly steps [133] and less likely to receive breastfeeding advice from their health professionals [134]. Investment to narrow the gap between the richest and the poorest is therefore vital.

Investment can and does work. Brazil, for example, is an excellent example of how implementing such a society wide approach significantly increases breastfeeding rates. In 1986, the median duration of breastfeeding was 2.5 months but, by 2006, it had risen to 14 months. Exclusive breastfeeding rates to 4 months also increased from 4% to 48% [135]. To undertake this, the government invested heavily in promoting breastfeeding at the societal level, including multi-organisation working, media campaigns, training for health workers and the development of mother-to-mother support groups. Policywise, a strict enforcement of the International Code was introduced, maternity leave was extended from to 6 months and more than 300 maternity hospitals gained Baby Friendly Hospital Initiative certification. Investment in over 200 human milk banks led to Brazil having the highest number in the world. These interventions were successful as a result of their combination, as well as the fact that they did not focus solely on maternal knowledge, instead focusing on a mother's wider environment and support system, enabling her to breastfeed her baby [136].

Tackling our low breastfeeding rates should therefore not only focus on fixing the physical issues that women eventually present with, but also should systematically break down the environmental factors that negatively impact upon breastfeeding attitudes, intentions and ability. Given that infant feeding decisions are not made in isolation, strategies to raise breastfeeding rates should not purely be individualistic. Although good quality, individualised one-to-one support is vital, it is unfair to hold women responsible for behaviour that is affected by structural factors. Morally, given the negative mental health impacts that a failed breastfeeding experience can bring, should we really be promoting breastfeeding unless we also provide an environment that is conducive to its success? We must hold public health services accountable to raising and sustaining breastfeeding rates rather than placing responsibility and blame in the laps of individual mothers[137].

Conflict of interests, source of funding and authorship

The author declares that she has no conflict of interest.
No funding declared.
AB was solely responsible for all aspects of this work.

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